

# Operator Algebra

## Linear Algebra

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Autumn 2026

Algebra of  $\text{End}(\mathcal{V})$ Functions of  
Operators

Commutators

Derivatives of  
Operators

Hermitian Operators

Direct Sums

Idempotents

Projection  
OperatorsRepresentation of  
Algebras

## Remark

*In the linear space  $\text{End}(\mathcal{V})$  we define the product as the composition of maps. In addition to the zero element, which is present in all algebras,  $\text{End}(\mathcal{V})$  has an identity element  $\mathbf{1}$ . Thus,  $\text{End}(\mathcal{V})$  is an unital algebra. With  $\mathbf{1}$  in the algebra, one may ask whether it is possible to find an operator  $\mathbf{T}^{-1}$  with the property that  $\mathbf{T}^{-1}\mathbf{T} = \mathbf{T}\mathbf{T}^{-1} = \mathbf{1}$ .*

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### Remark

*By the corollary following Theorem 1.17 (Dimension Theorem) we recall that an element of  $\text{End}(\mathcal{V})$  is bijective if and only if it is either injective or surjective.*

### Example 3.1

Let the linear operator  $\mathbf{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be defined by

$$\mathbf{T}(x) = (x_1 + x_2, x_2 + x_3, x_3 + x_1).$$

Determine whether  $\mathbf{T}$  is invertible and, if so, find its inverse.

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## Corollary 3.2

*The inverse of a linear operator is unique. If  $\mathbf{T}$  and  $\mathbf{S}$  are two invertible operators on  $\mathcal{V}$ , then  $\mathbf{TS}$  is also invertible and*

$$(\mathbf{TS})^{-1} = \mathbf{S}^{-1}\mathbf{T}^{-1}.$$

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### Theorem 3.3

*An endomorphism  $\mathbf{T} \in \text{End}(\mathcal{V})$  is invertible if and only if it sends a basis of  $\mathcal{V}$  onto another basis of  $\mathcal{V}$ .*

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## Remark

*Negative powers of an invertible operator are defined through  $\mathbf{T}^{-m} = (\mathbf{T}^{-1})^m$ . Later in the course (when we discuss the spectral decomposition theorem) we will generalise this idea to exponents of any real number.*

### Example 3.4

Let  $\mathbf{T}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the linear operator that rotates vectors in the  $xy$ -plane through an angle  $\theta$ , find  $\mathbf{T}_\theta^n$ . Further, check that  $\mathbf{T}$  is invertible and has inverse

$$\mathbf{T}_\theta^{-n} = \mathbf{T}_{-n\theta}.$$

## Remark

From the chapter on algebras, we know that we can construct polynomials of  $\mathbf{T} \in \text{End}(\mathcal{V})$  such as

$$p(\mathbf{T}) = \alpha_\mu \mathbf{T}^\mu \quad \mu = 1, \dots, n.$$

We can go one step beyond polynomials of operators and, via Taylor expansion, define functions of them. To do so, consider the Taylor series of a function  $f$

$$f(x) = \sum_{k=0}^{\infty} \frac{(x - x_0)^k}{k!} \left. \frac{d^k f}{dx^k} \right|_{x=x_0}$$

in which  $x_0$  is a point where  $f$  and all its derivatives are defined. To this function, there corresponds a function of the operator  $\mathbf{T}$ , defined as

$$f(\mathbf{T}) = \sum_{k=0}^{\infty} \left. \frac{d^k f}{dx^k} \right|_{x=x_0} \frac{(\mathbf{T} - x_0 \mathbf{1})^k}{k!}.$$

Because this series is an infinite sum of operators, difficulties may arise concerning its convergence. However, it can be shown that  $f(\mathbf{T})$  is always defined for finite-dimensional linear spaces. In fact, it is always a polynomial in  $\mathbf{T}$ . For the time being, we shall think of  $f(\mathbf{T})$  as a formal infinite series<sup>†</sup>. A simplification results when the function can be expanded about  $x = 0$ . In this case we obtain

$$f(\mathbf{T}) = \sum_{k=0}^{\infty} \left. \frac{d^k f}{dx^k} \right|_{x=0} \frac{\mathbf{T}^k}{k!}.$$

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<sup>†</sup>Despite the name, the concept of “formal infinite series” refers to the algebraic manipulation of infinite series avoiding questions of convergence.

## Example 3.5

Evaluate  $\exp(\alpha \mathbf{T})$  when  $\mathbf{T} \in \text{End}(\mathbb{R}^2)$  is given by

$$\mathbf{T}(x, y) = (-y, x).$$

Show that  $\exp(\alpha \mathbf{T})$  is a polynomial (of first degree) in  $\mathbf{T}$ .

## Definition

We can think of  $e^{\alpha \mathbf{T}}$  as a rotation operator of an angle  $\alpha$  about the  $z$ -axis. In this context  $\mathbf{T}$  is called the **generator** of the rotation.

## Definition

The **commutator**  $[\mathbf{U}, \mathbf{T}]$  of the two operators  $\mathbf{U}$  and  $\mathbf{T}$  in  $\text{End}(\mathcal{V})$  is another operator in  $\text{End}(\mathcal{V})$ , defined as

$$[\mathbf{U}, \mathbf{T}] = \mathbf{UT} - \mathbf{TU}.$$

### Theorem 3.6 (Properties of the commutator)

For  $\mathbf{A}, \mathbf{S}, \mathbf{T}, \mathbf{U} \in \text{End}(\mathcal{V})$  and  $\alpha, \beta \in \mathbb{C}$ , we have

- i) **(Antisymmetry)**  $[\mathbf{U}, \mathbf{T}] = -[\mathbf{T}, \mathbf{U}]$ .
- ii) **(Bilinearity)**  
 $[\alpha\mathbf{U} + \mathbf{S}, \beta\mathbf{T} + \mathbf{A}] = \alpha\beta([\mathbf{U}, \mathbf{T}] + [\mathbf{U}, \mathbf{A}] + [\mathbf{S}, \mathbf{T}] + [\mathbf{S}, \mathbf{A}])$ .
- iii) **(Right derivation property)**  $[\mathbf{ST}, \mathbf{U}] = \mathbf{S}[\mathbf{T}, \mathbf{U}] + [\mathbf{S}, \mathbf{U}]\mathbf{T}$ .
- iv) **(Left derivation property)**  $[\mathbf{S}, \mathbf{TU}] = \mathbf{T}[\mathbf{S}, \mathbf{U}] + [\mathbf{S}, \mathbf{T}]\mathbf{U}$ .
- v) **(Jacobi identity)**  $[[\mathbf{S}, \mathbf{T}], \mathbf{U}] + [[\mathbf{U}, \mathbf{S}], \mathbf{T}] + [[\mathbf{T}, \mathbf{U}], \mathbf{S}] = 0$ .

## Definition

For a mapping  $\mathbf{H} : (a, b) \rightarrow \text{End}(\mathcal{V})$  we define the **derivative** as

$$\frac{d\mathbf{H}(t_0)}{dt} = \lim_{h \rightarrow 0} \frac{\mathbf{H}(t_0 + h) - \mathbf{H}(t_0)}{h}, \quad t_0 \in (a, b).$$

### Example 3.7

*Find the derivative of  $\exp(t\mathbf{H})^\dagger$ , where  $\mathbf{H}$  is independent of  $t$ .*

<sup>†</sup>Here, you may assume that  $\mathbf{H}$  and  $\exp(t\mathbf{H})$  commute, and that  $\exp(t\mathbf{H})\exp(h\mathbf{H}) = \exp((t+h)\mathbf{H})$ .

## Example 3.8

Consider the Schrödinger equation

$$i \partial_t |\psi(t)\rangle = \mathbf{H} |\psi(t)\rangle .$$

Define the so-called **evolution operator**  $\mathbf{U}(t)$  by  $|\psi(t)\rangle = \mathbf{U}(t) |\psi(0)\rangle$ , and thus show that the solution to Schrödinger's equation can be written as

$$|\psi(t)\rangle = \exp(-it\mathbf{H}) |\psi(0)\rangle$$

## Lemma 3.9

If  $[\mathbf{S}, \mathbf{T}]$  commutes with  $\mathbf{S}$  and  $\mathbf{T}$ , then  $[e^{t\mathbf{T}}, \mathbf{S}] = -t[\mathbf{S}, \mathbf{T}]e^{t\mathbf{T}}$ .

### Theorem 3.10 (Baker-Campbell-Hausdorff Formula)

Let  $\mathbf{S}, \mathbf{T} \in \mathcal{L}(\mathcal{V})$ . If  $[\mathbf{S}, [\mathbf{S}, \mathbf{T}]] = 0 = [\mathbf{T}, [\mathbf{S}, \mathbf{T}]]$ , then the **Baker-Campbell-Hausdorff formula** holds:

$$\exp(t\mathbf{S}) \exp(t\mathbf{T}) \exp(-t^2/2[\mathbf{S}, \mathbf{T}]) = \exp(t(\mathbf{S} + \mathbf{T})).$$

In particular,  $\exp(t\mathbf{S}) \exp(t\mathbf{T}) = \exp(t(\mathbf{S} + \mathbf{T}))$  if and only if  $[\mathbf{S}, \mathbf{T}] = 0$ .

## Definition

The **anticommutator**  $\{\mathbf{S}, \mathbf{T}\}$  of the operators  $\mathbf{S}$  and  $\mathbf{T}$  is defined as

$$\{\mathbf{S}, \mathbf{T}\} = \mathbf{ST} + \mathbf{TS}.$$

## Theorem 3.11

Let  $\mathbf{H} : \mathbb{R} \rightarrow \mathcal{L}(\mathcal{V})$  and assume that  $\mathbf{H}$  and its derivative commute with  $[\mathbf{H}, d\mathbf{H}/dt]$ . Then

$$\frac{d}{dt} \exp(\mathbf{H}(t)) = \frac{1}{2} \left\{ \frac{d\mathbf{H}}{dt}, \exp(\mathbf{H}) \right\}.$$

In particular, if  $[\mathbf{H}, d\mathbf{H}/dt] = 0$ , then

$$\frac{d}{dt} e^{\mathbf{H}(t)} = \frac{d\mathbf{H}}{dt} e^{\mathbf{H}} = e^{\mathbf{H}} \frac{d\mathbf{H}}{dt}.$$

## Example 3.12

Define  $\mathbf{F} \doteq e^{t\mathbf{A}}\mathbf{B}e^{-t\mathbf{A}}$  and let  $\mathbf{A}, \mathbf{B}$  be  $t$ -independent. Show that

$$\frac{d\mathbf{F}}{dt} = [\mathbf{A}, \mathbf{F}(t)] \quad \text{and} \quad \frac{d}{dt} [\mathbf{A}, \mathbf{F}(t)] = \left[ \mathbf{A}, \frac{d\mathbf{F}}{dt} \right].$$

Using this result, show that

$$e^{t\mathbf{A}}\mathbf{B}e^{-t\mathbf{A}} = \left( \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{A}^n \right) [\mathbf{B}] \equiv \mathbf{B} + t[\mathbf{A}, \mathbf{B}] + \frac{t^2}{2!} [\mathbf{A}, [\mathbf{A}, \mathbf{B}]] + \cdots \equiv e^{t\mathbf{A}}[\mathbf{B}].$$

Express this result for the case  $t = 1$  and check what happens when  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ .

## Definition

Let  $\mathbf{T} \in \text{End}(\mathcal{V})$  and  $|a\rangle, |b\rangle \in \mathcal{V}$ , The **hermitian conjugate**, (or **adjoint**), of  $\mathbf{T}$  is denoted by  $\mathbf{T}^\dagger$  and defined by

$$\langle a | \mathbf{T} | b \rangle^* = \langle b | \mathbf{T}^\dagger | a \rangle .$$

Whence, we identify

$$\langle c | = \langle b | \mathbf{T}^\dagger \implies (\mathbf{T} | b \rangle)^\dagger = \langle b | \mathbf{T}^\dagger .$$

*This equation is sometimes used as the definition of hermitian conjugate.*

## Theorem 3.13

Let  $\mathbf{U}, \mathbf{T} \in \text{End}(\mathcal{V})$  and  $\alpha \in \mathbb{C}$ . Then

1.  $(\mathbf{U} + \mathbf{T})^\dagger = \mathbf{U}^\dagger + \mathbf{T}^\dagger.$
2.  $(\mathbf{UT})^\dagger = \mathbf{T}^\dagger \mathbf{U}^\dagger.$
3.  $(\alpha \mathbf{T})^\dagger = \alpha^* \mathbf{T}^\dagger.$
4.  $\left((\mathbf{T})^\dagger\right)^\dagger = \mathbf{T}.$

Where the last equality does not hold in infinite-dimensional linear spaces.

## Remark

*In previous examples dealing with linear operators  $\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ , an element of  $\mathbb{R}^n$  was denoted by a row vector, such as  $(x, y)$  for  $\mathbb{R}^2$  and  $(x, y, z)$  for  $\mathbb{R}^3$ . There was no confusion, because we were operating only in  $\mathcal{V}$ . However, since elements of both  $\mathcal{V}$  and  $\mathcal{V}^*$  are required when discussing  $\mathbf{T}, \mathbf{T}^*$ , and  $\mathbf{T}^\dagger$ , it is helpful to make a distinction between them. We therefore introduce the convention that kets are represented as column vectors and bras as row vectors.*

### Example 3.14

Find the hermitian conjugate of the operator  $\mathbf{T} : \mathbb{C}^3 \rightarrow \mathbb{C}^3$  given by

$$\mathbf{T} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} \alpha_1 - i\alpha_2 + \alpha_3 \\ i\alpha_1 - \alpha_3 \\ \alpha_1 - \alpha_2 + i\alpha_3 \end{pmatrix}.$$

## Definition

A linear operator  $\mathbf{H} \in \text{End}(\mathcal{V})$  is called **hermitian** (or **self-adjoint**) if  $\mathbf{H}^\dagger = \mathbf{H}$ .  
Similarly,  $\mathbf{A} \in \text{End}(\mathcal{V})$  is called **anti-hermitian** if  $\mathbf{A}^\dagger = -\mathbf{A}$ .



Charles Hermite  
1822-1901

## Historical Notes

**Charles Hermite**, one of the most eminent French mathematicians of the nineteenth century, was particularly distinguished for the clean elegance and high artistic quality of his work. As a student, he courted disaster by neglecting his routine assigned work to study the classic masters of mathematics; and though he nearly failed his examinations, he became a first-rate creative mathematician while still in his early twenties. In 1870 he was appointed to a professorship at the Sorbonne, where he trained a whole generation of well-known French mathematicians, including Picard, Borel, and Poincaré. His 1873 proof of the transcendence of  $e$  was another high point of his career. If he had been willing to dig even deeper into this vein, he could probably have disposed of  $\pi$  as well, but apparently he had enough of a good thing. As he wrote to a friend, “I shall risk nothing on an attempt to prove the transcendence of the number  $\pi$ . If others undertake this enterprise, no one will be happier than I at their success, but believe me, my dear friend, it will not fail to cost them some efforts.” As it turned out, Lindemann’s proof nine years later rested on extending Hermite’s method. Several of his purely mathematical discoveries had unexpected applications many years later to mathematical physics. For example, the Hermitian forms and matrices that he invented in connection with certain problems of number theory turned out to be crucial for Heisenberg’s 1925 formulation of quantum mechanics, and Hermite polynomials are useful in solving Schrödinger’s wave equation.

## Definition

The **expectation value**  $\langle \mathbf{T} \rangle_a$  of an operator  $\mathbf{T}$  in the “state”  $|a\rangle$  is a complex number defined by  $\langle \mathbf{T} \rangle_a = \langle a | \mathbf{T} | a \rangle$ . When no risk of confusion exists, it is common to drop the subscript “ $a$ ” and write  $\langle \mathbf{T} \rangle$  for the expectation value.

## Remark

*Any operator can be written as a sum of a hermitian operator and an anti-hermitian operator. The decomposition is given by*

$$\mathbf{T} = \frac{1}{2}(\mathbf{T} + \mathbf{T}^\dagger) + \frac{1}{2}(\mathbf{T} - \mathbf{T}^\dagger) \equiv \mathbf{X} + \mathbf{A}.$$

*Furthermore, if we let  $\mathbf{Y} \doteq -i\mathbf{A}$ , then*

$$\mathbf{T} = \mathbf{X} + i\mathbf{Y}$$

*where  $\mathbf{X}$  and  $\mathbf{Y}$  are both hermitian.*

## Theorem 3.15

*A linear map  $\mathbf{H}$  on a complex inner product space is hermitian if and only if  $\langle a | \mathbf{H} | a \rangle$  is real for all  $|a\rangle$ .*

**Example 3.16**

Show that any operator  $\mathbf{H} \in \text{End}(\mathbb{C}^2)$  of the form

$$\mathbf{H} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} \alpha z_1 + \beta z_2 \\ \beta^* z_1 + \gamma z_2 \end{pmatrix}$$

with  $\alpha, \gamma \in \mathbb{R}, \beta \in \mathbb{C}$ , is hermitian.

## Definition

An operator  $\mathbf{A}$  on an inner product space is called **nonnegative** (written  $\mathbf{A} \geq 0$ ) if  $\langle a | \mathbf{A} | a \rangle \geq 0$  for all  $|a\rangle \neq |0\rangle$ . We say  $\mathbf{A}$  is **positive definite** (written  $\mathbf{A} > 0$ ) if  $\langle a | \mathbf{A} | a \rangle > 0$  for all  $|a\rangle \neq |0\rangle$ .

## Corollary 3.17

*A positive definite operator is invertible.*

## Example 3.18

*Show that the square of a hermitian operator is a nonnegative operator. Moreover, show that the square of an invertible hermitian operator is positive definite.*

## Definition

Let  $\mathcal{V}$  be a finite-dimensional inner product space. An operator  $\mathbf{U}$  is called a **unitary operator** if  $\mathbf{U}^\dagger = \mathbf{U}^{-1}$ . Unitary operators preserve the inner product of  $\mathcal{V}$ .

## Example 3.19

Show that the linear operator  $\mathbf{T} \in \text{End}(\mathbb{C}^3)$  given by

$$\mathbf{T} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} (z_1 - iz_2)/\sqrt{2} \\ (z_1 + iz_2 - 2z_3)/\sqrt{6} \\ [z_1 - z_2 + z_3 + i(z_1 + z_2 + z_3)]/\sqrt{6} \end{pmatrix}$$

is unitary.

## Definition

*Let  $\mathcal{U}$  and  $\mathcal{W}$  be subspaces of a linear space  $\mathcal{V}$ . We denote by  $\mathcal{U} + \mathcal{W}$  the collection of all vectors in  $\mathcal{V}$  that can be written as a sum of two vectors, one in  $\mathcal{U}$  and one in  $\mathcal{W}$ .*

## Example 3.20

*Let  $\mathcal{U}$  be the  $xy$ -plane and  $\mathcal{W}$  the  $yz$ -plane. These are both subspaces of  $\mathbb{R}^3$ , and so is  $\mathcal{U} + \mathcal{W}$ . In fact, show that  $\mathcal{U} + \mathcal{W} = \mathbb{R}^3$ .*

## Definition

Let  $\mathcal{U}$  and  $\mathcal{W}$  be subspaces of a linear space  $\mathcal{V}$  such that  $\mathcal{V} = \mathcal{U} + \mathcal{W}$  and  $\mathcal{U} \cap \mathcal{W} = \{0\}$ . Then we say that  $\mathcal{V}$  is the **direct sum** of  $\mathcal{U}$  and  $\mathcal{W}$  and write  $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ .

**Theorem 3.21**

*Let  $\mathcal{U}$  and  $\mathcal{W}$  be subspaces of  $\mathcal{V}$  such that  $\mathcal{V} = \mathcal{U} + \mathcal{W}$ . Then  $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$  if and only if any nonzero vector in  $\mathcal{V}$  can be written uniquely as a vector in  $\mathcal{U}$  plus a vector in  $\mathcal{W}$ .*

**Definition**

Let  $\{\mathcal{U}\}_{i=1}^r$  be subspaces of  $\mathcal{V}$  such that

$$\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_r \quad \text{and} \quad \mathcal{U}_i \cap \mathcal{U}_j = \{0\} \quad \text{for all } i, j = 1, \dots, r.$$

Then we say that  $\mathcal{V}$  is the direct sum of  $\{\mathcal{U}_i\}_{i=1}^r$  and write

$$\mathcal{V} = \bigoplus_{i=1}^r \mathcal{U}_i.$$

## Proposition 3.22

*The vectors in different subspaces from the previous definition are linearly independent.*

**Proposition 3.23**

*Let  $\mathcal{U}$  be a subspace of  $\mathcal{V}$ . Then there exists a subspace  $\mathcal{W}$  of  $\mathcal{V}$  such that  $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ .*

## Theorem 3.24

*If  $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ , then  $\dim \mathcal{V} = \dim \mathcal{U} + \dim \mathcal{W}$ .*

**Theorem 3.25**

Let  $\mathcal{V}$  be the direct sum of  $\{\mathcal{U}_j\}$  and let  $\mathbf{P}_j \in \text{End}(\mathcal{V})$  be defined by  $\mathbf{P}_j |v\rangle = |v_j\rangle$  with  $|v_j\rangle$  the component of  $|v\rangle$  in  $\mathcal{U}_j$ . Then  $\{\mathbf{P}_i\}_{i=1}^r$  is a complete set of orthogonal idempotents, i.e.,

$$\mathbf{P}_i \mathbf{P}_j = \delta_{ij} \mathbf{P}_i \text{ (no sum)} \quad \text{and} \quad \sum_{j=1}^r \mathbf{P}_j = \mathbf{1}.$$

**Theorem 3.26**

If  $\mathbf{T} \in \text{End}(\mathcal{V})$  and  $\{\mathbf{P}_i\}_{i=1}^r$  is a complete set of orthogonal idempotents corresponding to  $\{\mathcal{U}_i\}_{i=1}^r$ , then  $\mathbf{T}$  can be written as a sum of operators which restrict to the subspaces. More precisely,

$$\mathbf{T} = \sum_{j=1}^r \mathbf{T}_j \quad \text{with} \quad \mathbf{T}_j = \mathbf{P}_j \mathbf{T}$$

where  $\mathbf{T}_j \in \text{End}(\mathcal{U}_j)$ .

## Definition

*A hermitian idempotent of  $\text{End}(\mathcal{V})$  is called a **projection operator**.*

## Theorem 3.27

Let  $\mathbf{P}_1, \mathbf{P}_2 \in \text{End}(\mathcal{V})$  be projection operators. Then  $\mathbf{P}_1 + \mathbf{P}_2$  is a projection operator if and only if  $\mathbf{P}_1$  and  $\mathbf{P}_2$  are orthogonal. More generally, if there is a set  $\{\mathbf{P}_i\}_{i=1}^m$  of projection operators satisfying  $\mathbf{P}_i \mathbf{P}_j = \delta_{ij} \mathbf{P}_i$  (no sum), then

$$\mathbf{P} = \sum_{i=1}^m \mathbf{P}_i$$

is also a projection operator.

## Theorem 3.28

*Let  $|e\rangle$  be any normalised vector  $|e\rangle$  in an inner product space  $\mathcal{V}$ . Then  $\mathbf{P} = |e\rangle \langle e|$  is a projection operator.*

## Definition

For any nonzero vector  $|y\rangle$ , the projection operator  $\mathbf{P}_y$  along  $|y\rangle$  and the reflection operator  $\mathbf{R}_y$  in the plane perpendicular to  $|y\rangle$  are given by

$$\mathbf{P}_y = \frac{|y\rangle \langle y|}{\langle y|y\rangle} \quad \text{and} \quad \mathbf{R}_y = \mathbf{1} - 2\mathbf{P}_y = \mathbf{1} - 2\frac{|y\rangle \langle y|}{\langle y|y\rangle}.$$

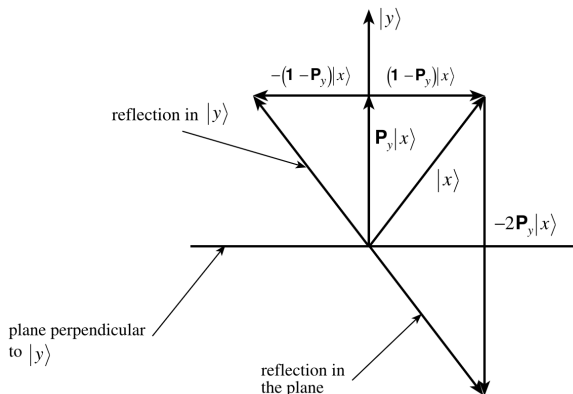


Figure 1: Graphical representation of  $\mathbf{P}_y$  and  $\mathbf{R}_y$

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## Theorem 3.29

Let  $B = \{|e_i\rangle\}_{i=1}^N$  be an orthonormal basis for  $\mathcal{V}_N$ . Then the set  $\{\mathbf{P}_i = |e_i\rangle\langle e_i|\}_{i=1}^N$  consists of mutually orthogonal projection operators, and

$$\sum_{i=1}^N \mathbf{P}_i = \sum_{i=1}^N |e_i\rangle\langle e_i| = \mathbf{1}.$$

This relation is called the **completeness relation**.

**Remark**

*If we choose only the first  $m < N$  vectors, then the projection operator*

$$\mathbf{P}^{(m)} \doteq \sum_{i=1}^m |e_i\rangle \langle e_i|$$

*projects arbitrary vectors into the subspace spanned by the first  $m$  basis vectors  $\{|e_i\rangle\}_{i=1}^m$ . In other words, when  $\mathbf{P}^{(m)}$  acts on any vector  $|a\rangle \in \mathcal{V}$ , the result will be a linear combination of only the first  $m$  vectors. These points are illustrated in the following example.*

## Example 3.30

Consider orthonormal basis  $\{|e_i\rangle\}_{i=1}^3 \in \mathbb{R}^3$  given by

$$|e_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad |e_2\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad |e_3\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

Construct  $\mathbf{P}_2$ , and by projecting an arbitrary vector  $|x\rangle \in \mathbb{R}^3$  with  $\mathbf{P}_2$  and then computing  $\langle e_1 | \mathbf{P}_2 | x \rangle$ ,  $\langle e_3 | \mathbf{P}_2 | x \rangle$ , show that  $\mathbf{P}_2$  projects onto the line along  $|e_2\rangle$ . Moreover, find the operator  $\mathbf{P}_{1,2}$  that projects onto the plane formed by  $|e_1\rangle$  and  $|e_2\rangle$ . Verify your result by showing that  $\langle e_3 | \mathbf{P}_{1,2} | x \rangle = 0$  for any  $|x\rangle \in \mathbb{R}^3$ . Further, verify that the completeness relation holds.

## Example 3.31

Consider a vector

$$|y\rangle = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}.$$

Find the most general projection and reflection operator in  $\mathbb{R}^2$ . Use this result to show that the product of two reflections in  $\mathbb{R}^2$  is a rotation, and in particular, an isometry.

**Theorem 3.32**

*Let  $\mathcal{V}$  be an  $N$ -dimensional inner product space. Then any isometry  $\mathbf{T}$  of  $\mathcal{V}$  is the product of at most  $N + 1$  reflections.*

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## Definition

*We will write both  $\mathbb{F}$  and  $\mathbb{K}$  to denote either  $\mathbb{R}$  or  $\mathbb{C}$  with the condition that  $\mathbb{F} \subset \mathbb{K}$ . If  $\mathcal{V}$  is a linear space over  $\mathbb{K}$ , we denote the algebra of its endomorphisms by  $\text{End}_{\mathbb{K}}(\mathcal{V})$ . When there is no risk of confusion, we remove the subscript  $\mathbb{K}$ .*

## Definition

Let  $\mathcal{A}$  be an associative algebra over  $\mathbb{F}$  with identity  $\mathbf{1}_{\mathcal{A}}$ . A  **$\mathbb{K}$ -representation** of  $\mathcal{A}$  in a linear space  $\mathcal{V}$  over  $\mathbb{K}$  is a homomorphism  $\rho : \mathcal{A} \rightarrow \text{End}_{\mathbb{K}}(\mathcal{V})$  such that  $\rho(\mathbf{1}_{\mathcal{A}}) = \mathbf{1}$ , where  $\mathbf{1}$  is the unit operator in  $\text{End}_{\mathbb{K}}(\mathcal{V})$ . The representation  $\rho$  is said to be **faithful** if it is injective.

### Example 3.33

Let  $\mathcal{A} = \mathbb{C}$  be the algebra of complex numbers, i.e., elements of the form

$$z = x^j \mathbf{e}_j, \quad j = 1, 2, \quad x^j \in \mathbb{R}.$$

Let  $\mathcal{V} = \mathbb{R}^2$  and consider  $\rho : \mathbb{C} \rightarrow \text{End}_{\mathbb{R}}(\mathbb{R}^2)$ , the real representation of complex numbers, given by

$$\rho(z) = \begin{pmatrix} x^0 & -x^1 \\ x^1 & x^0 \end{pmatrix}.$$

Use Proposition 2.11 to show that  $\rho$  is a faithful representation.

### Example 3.34

Let  $\mathcal{A} = \mathbb{H}$  be the (real) algebra of quaternions, i.e., elements of the form

$$q = q^\mu \mathbf{e}_\mu, \quad \mu = 1, 2, 3, 4, \quad q^\mu \in \mathbb{R}.$$

Let  $\mathcal{V} = \mathbb{R}^4$  and consider  $\rho : \mathbb{H} \rightarrow \text{End}_{\mathbb{R}}(\mathbb{R}^4)$ , the real representation of quaternions, given by

$$\rho(q) = \begin{pmatrix} q^0 & -q^1 & -q^2 & -q^3 \\ q^1 & q^0 & -q^3 & q^2 \\ q^2 & q^3 & q^0 & -q^1 \\ q^3 & -q^2 & q^1 & q^0 \end{pmatrix}.$$

Use Proposition 2.11 to show that  $\rho$  is a faithful representation.